

Unidirectional Local-Attention Autoencoder Network for Spectral Variability Unmixing

Shu Xiang^{ID}, Xiaorun Li^{ID}, Jigang Ding^{ID}, Shuhan Chen^{ID}, *Member, IEEE*, and Ziqiang Hua^{ID}

Abstract—Autoencoders (AEs) have demonstrated excellent performance in the field of hyperspectral unmixing (SU), due to their self-supervised nature and ease of implementation. Recently proposed AE-based networks contend that local spatial information limits further improvement in unmixing accuracy and tends to explore and utilize global information, which improves unmixing accuracy at the expense of increased computational complexity. However, we believe that precise unmixing can be achieved by fully leveraging local information. In this article, we propose a unidirectional local-attention AE network (ULA-Net) that explores spatial information pixel by pixel and achieves accurate spatial-spectral feature fusion. ULA-Net utilizes unidirectional local attention (ULA) module to calculate the correlation between neighboring pixels and the central pixel within local regions, extracting discriminative local information. Moreover, ULA-Net effectively extracts relevant spatial information and suppresses irrelevant information based on a double fusion strategy (DFS) module. This process achieves more accurate control over the contribution of spatial information by implementing information fusion in both the pixel and feature dimensions. To address spectral variability, we implement the extended linear mixing model (ELMM) in the decoder part to improve unmixing accuracy without increasing the number of parameters. We conduct ablation experiments to investigate the roles of each module. Experimental results on both synthetic and real datasets demonstrate the effectiveness of the proposed network.

Index Terms—Autoencoder (AE) network, hyperspectral unmixing, multihead attention mechanism, spectral variability, spectral-spatial model.

I. INTRODUCTION

HYPERSPECTRAL unmixing [1] is a technique that decomposes a pixel into a set of material spectra, called endmembers, and their fractions in the pixel, called abundances. In a remote sensing scene, far imaging distance and limited spatial resolution of the imaging device result in mixed pixels, which means several materials are captured in one-pixel [2]. Fortunately, due to the abundant spectral information of a hyperspectral image (HSI), each pixel contains the

signatures that can be used to recover the ground distribution by spectral unmixing (SU) approaches.

Based on the reflection mechanism of photons in physics, SU is designed to invert the mixing process to disentangle the substances in one pixel. The most commonly used model is the linear mixing model (LMM), which is concise and efficient. It assumes that the simple aggregation of material spectra composes the mixed spectrum, which implies the irrelevancy of spatial pixels and the independence of items in pixels. In real-world scenarios, the spectral signatures of pure constituent materials exhibit variations across the observed HSI. This phenomenon is referred to as spectral variability, typically induced by variations in illumination and diverse atmospheric conditions [3], [4], [5]. In real scenes, achieving the ideal conditions assumed by LMM can be challenging, which causes modeling errors between the estimation and physical truth.

Unmixing considering spectral variability offers one approach to address this issue [2], [6], [7]. Typically, unmixing methods for addressing spectral variability can be categorized into three classes: endmember bundles-based methods [8], [9], [10], probability distribution-based methods [11], [12], [13], and modified LMMs (MLMMs) [14], [15], [16]. Endmember bundles-based approaches tackle spectral variability by representing the same land cover with a finite number of spectral signatures. Probability distribution-based methods model spectral variability in real-world scenes by imposing different probability distributions on endmember, endmember variability, and abundance. MLMMs account for the impact of spectral variability on real-world scenes, considering factors such as scaling factor and perturbation term. MLMMs have gained much attention due to their advantages of interpretability, small number of parameters, and explicit consideration for spectral variability.

The above models are solved utilizing conventional optimization algorithms, such as convex optimization, expectation-maximum (EM), and Markov chain Monte Carlo (MCMC) algorithms [17]. In recent years, the deep learning-based optimization method by applying neural networks as the framework in the field of SU has attracted extensive attention from researchers [18], [19], [20]. The autoencoder (AE), with its end-to-end learning capability, can be applied to the unsupervised unmixing of HSIs [21]. In AEs, the encoder outputs are considered as abundances, while the weights of the decoder represent endmembers.

To leverage the latent deep features embedded in HSIs, it is crucial to design an encoder that efficiently explores

Manuscript received 7 November 2023; revised 16 January 2024 and 20 February 2024; accepted 7 March 2024. Date of publication 14 March 2024; date of current version 21 March 2024. The work of Shuhan Chen was supported in part by the National Natural Science Foundation of China under Grant 62171404 and in part by the Joint Fund Project of the Chinese Ministry of Education under Grant 8091B022118. (*Corresponding author: Shuhan Chen.*)

Shu Xiang, Xiaorun Li, Jigang Ding, and Shuhan Chen are with the College of Electrical Engineering, Zhejiang University, Hangzhou 310027, China (e-mail: 12210076@zju.edu.cn; lxr@zju.edu.cn; 12010075@zju.edu.cn; 11410057@zju.edu.cn).

Ziqiang Hua is with the Intelligent Network Research Institute, Zhejiang Laboratory, Hangzhou 311121, China (e-mail: ziqhua@zhejianglab.com).

Digital Object Identifier 10.1109/TGRS.2024.3375598

useful information while mitigating irrelevant interference. The design of encoders plays an essential role in achieving effective unmixing. The encoder of uDAS [18] consists of a single FC layer, while DAEN [20] employs a stacked AE (SAE) to initialize the variational AE (VAE) with learned spectral features. In contrast to imposing a prior to models, DeepGUN [19] leverages a VAE to learn spectral variability features, enhancing unmixing performance. From the perspective of spectral information, these algorithms have explored the spectral variability of HSI to suppress irrelevant information and improve unmixing accuracy.

However, extensive research has shown that incorporating neighborhood spatial information of pixels can further enhance unmixing performance [22]. CNNAEU [23] utilizes a convolution AE to leverage spatial information. EGU-Net [24], built upon the concept of endmember-guided networks, substantiates that the incorporation of spatial information within the unmixing network yields a higher level of accuracy. Zhao et al. [25] introduced 3-D CNN to extract spatial and spectral information simultaneously. Considering that over-reliance on spatial information could lead to overfitting, GTCAN [26] employs a gating mechanism to adjust the weight of spatial information adaptively. By employing gate regularization, it achieved a better balance between spectral and spatial information. CNN is an effective tool for feature extraction and fusion, but it may lack the ability to extract features with specific adaptability to different regions and may introduce irrelevant information and noise due to its weight-sharing property.

Considering that CNN networks cannot control the contribution of pixels during unmixing effectively, Attention mechanisms provide a solution for precise control over neighboring information. Attention mechanisms can assign weights to input pixels, allowing focused attention on relevant pixels. This enables better utilization of relevant information to accomplish the task. Multimodal-HU [27] and SWAU-Net [28] enhance unmixing accuracy by employing linear layers or convolutions to generate attention maps from features.

Transformers [29], [30], [31], with their self-attention mechanism, offer a better solution. Unlike traditional attention mechanisms, which rely on learned weights to generate attention maps, transformers calculate similarity coefficients between each pixel and other pixels using self-attention, achieving better performance in vision tasks. Recently, the self-attention mechanism in transformers has been proven effective in processing HSIs due to its correlation-based global attention computation. It has achieved reliable results in image classification [32], band selection [33], and change detection [34]. In the field of unmixing, DeepTrans [35] was the first to introduce the self-attention mechanism in vision-transformer, capturing contextual information by computing global attention on latent variables. Both UnDAT [36] and SSABN [37] utilize a window-based self-attention approach in the Swin transformer, simultaneously reducing computational complexity while computing global attention.

In order to effectively address the challenge of spectral variability, the design of the decoder is also important.

A single FC layer is no longer adequate for unmixing complex real-world scenarios. PGMSU [38] tackles endmember variability by utilizing a VAE to generate endmembers for each pixel. An AE considering additive nonlinear mixing [39] employs a dual-branch decoder to fit linear and additive nonlinear spectral mixing. Recently, there has been a growing interest in decoder designs based on MLMs. 3DCNN-var [25], designed upon the PLMM [14] model, constructs a decoder and utilizes variational inference to generate perturbation terms. MAAENet [40] trains a learnable parameter as scaling factors, achieving unmixing based on the ELMM [16].

A. Motivation

- 1) HSIs often exhibit local homogeneity [41]. Also, spectral variability caused by illumination and terrain often involves multiple pixels within a partial region. Thus, the affected pixels typically exhibit local similarity. Proper utilization of local contextual features can effectively improve unmixing performance. However, existing Transformer-based self-attention mechanisms are characterized by global attention. Global attention inevitably leads to information redundancy. The excessive information tends to perturb crucial details during feature fusion, resulting in unstable unmixing accuracy. In this article, we design a local attention mechanism built upon the self-attention mechanism that effectively exploits spatial information in local regions to enhance unmixing precision.
- 2) Self-attention mechanisms in transformers effectively enhance useful information while suppressing irrelevant information. This is accomplished by mapping the data into a feature space, and then computing the correlation between pixels to generate an attention map, which is applied to the respective pixels. Thus, the self-attention mechanism takes precise control over the contributions of every pixel. However, due to the Softmax operation in attention mechanisms, the attention map tends to be dense. In applications, even spatial information that may seem unrelated is sometimes utilized in the unmixing process. Consequently, it is essential to design a feature selection and fusion mechanism that further filters useful information and mitigates the impact of irrelevant data.
- 3) Among numerous decoders considering spectral variability, [39] utilizes extracted features to generate perturbation terms based on PLMM. However, due to the large number of parameters in PLMM, the model becomes challenging to train, and the perturbation terms may interfere with the accuracy of abundance estimation. To mitigate the ill-posedness caused by a large number of parameters, MAAENet incorporates a learnable parameter as a scaling factor based on the ELMM. Owing to the powerful representative capacity of the network, the model may be unstable to noise, leading to potential overfitting issues. Hence, how to design a decoder considering spectral variability and robustness to noise is one of the concerns of our algorithm.

B. Contributions

To address the issues mentioned above, we propose a unidirectional local-attention AE network (ULA-Net). The ULA module in our work leverages the frequently overlooked advantage of Transformers in exploring local information in a unidirectional way. This module facilitates feature extraction at the pixel level, ensuring accurate and detailed analysis. To mitigate the potential misuse of spatial information caused by dense attention maps, we propose a double fusion strategy (DFS). By implementing a double information fusion operation in both the pixel and feature dimensions, we aim to achieve a sparse attention map. This approach enables precise control over the contribution of each pixel, effectively suppressing noise and improving unmixing accuracy. In order to explicitly model spectral variability and mitigate overfitting, we introduce a scaling decoder, enabling the implementation of the ELMM in the AE framework. This scaling decoder provides a more flexible solution for AE-based unmixing networks to deal with spectral variability. To be specific, the main contribution of this article can be summarized as follows.

- 1) We propose a novel AE-based unmixing model called ULA-Net that effectively explores and leverages discriminative local spatial-spectral information while modeling spectral variability. In the field of unmixing, we introduce the attention mechanism of Transformers in local information exploration. The proposed network effectively resolves the challenges of high computational complexity and slow convergence encountered when handling global information and improves the accuracy of unmixing.
- 2) A ULA module is designed to unlock the potential of attention mechanisms in effectively exploring local information. This module is built in a unidirectional way to compute the correlation between neighboring pixels and the central pixel, enabling individual assessment of the significance of spatial pixels. By incorporating the ULA module into our framework, we enhance the ability to extract features in diverse patterns across different regions.
- 3) We propose a DFS module to alleviate the influence of irrelevant information on unmixing accuracy. This module efficiently regulates the influence of spatial information by incorporating a gate mechanism during feature fusion, resulting in a sparser attention map. This allows for flexible and precise control over local information.
- 4) We proposed an implementation method of ELMM in the AE to deal with spectral variability and avoid increasing the parameter count. The proposed scaling decoder exploits spectral variability information and generates scaling factors from extracted features to avoid potential overfitting.

The rest of this article is organized as follows. Section II provides an introduction to related works, including spectral mixing models, AE-based unmixing networks, and gated linear units (GLUs). In Section III, we present a detailed description

of our proposed ULA-Net, which comprises a ULA module, a DFS module, an abundance encoder, and a scaling decoder. Section IV discusses experimental results and provides comparisons with state-of-the-art algorithms. Finally, Section V concludes the article.

II. RELATED WORKS

In this section, related works of the proposed method will be introduced.

A. Spectral Mixing Models

1) *Linear Mixing Model*: Four components make up the basic LMM, they are the reshaped HSI ($\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3, \dots, \mathbf{y}_N] \in \mathbb{R}^{L \times N}$), endmembers ($\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \mathbf{m}_3, \dots, \mathbf{m}_P] \in \mathbb{R}^{L \times P}$), abundances ($\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \dots, \mathbf{a}_N] \in \mathbb{R}^{P \times N}$), and noises ($\mathbf{N} \in \mathbb{R}^{L \times N}$). The L , N , and P , respectively, denote the number of bands, pixels, and endmembers. The LMM in a matrix format can be described as

$$\begin{aligned} \mathbf{Y} &= \mathbf{MA} + \mathbf{N} \\ \text{s.t. } \mathbf{1}_P^T \mathbf{A} &= \mathbf{1}_N^T, \quad \mathbf{A} \geq 0, \mathbf{M} \geq 0 \end{aligned} \quad (1)$$

where the dot product of the endmember matrix and abundance matrix reconstruct the image matrix with noises. Since the model is physically explicable, three constraints should be satisfied. The abundance and endmember are nonnegative, which means reflectance and proportion have no negative values. Besides, the sum of the abundance fractions in one pixel should be 1.

2) *Extended LMM (ELMM)*: Based on the LMM, the ELMM addresses spectral variability from the perspective of spectral scaling. $\mathbf{S}_i \in \mathbb{R}^{P \times P}$ is the scaling term responsible for scaling the endmembers for each pixel. $\mathbf{S}_i$ is a diagonal matrix, and the values along the diagonal represent the scaling magnitudes applied to the P endmembers. The formula for ELMM is as follows:

$$\begin{aligned} \mathbf{y}_i &= \mathbf{MS}_i \mathbf{a}_i + \mathbf{n}_i \\ \text{s.t. } \mathbf{1}_P^T \mathbf{a}_i &= 1, \quad \mathbf{a}_i \geq 0, \mathbf{M} \geq 0, \mathbf{S}_i \geq 0 \end{aligned} \quad (2)$$

where $\mathbf{S}_i$ represents the scaling magnitude, its value in the parameter space must be greater than 0. When no scaling is required in the model, the default values along the diagonal of $\mathbf{S}_i$ are set to 1. In situations where the illumination intensity is higher in a particular region, the values of $\mathbf{S}_i$ are more likely to exceed 1, while conversely, they are less than 1 in areas with lower illumination intensity. $\mathbf{M}_i$ represents the fine-tuned endmembers for the i th pixel.

B. AE-Based Unmixing Networks

AEs are well suited for hyperspectral unmixing thanks to their flexibility and end-to-end training. Integrating AEs with other deep learning architectures also allows better modeling of mixed pixels and improves the separation of pure endmembers. As an emerging technique, AEs show great promise and potential in hyperspectral unmixing.

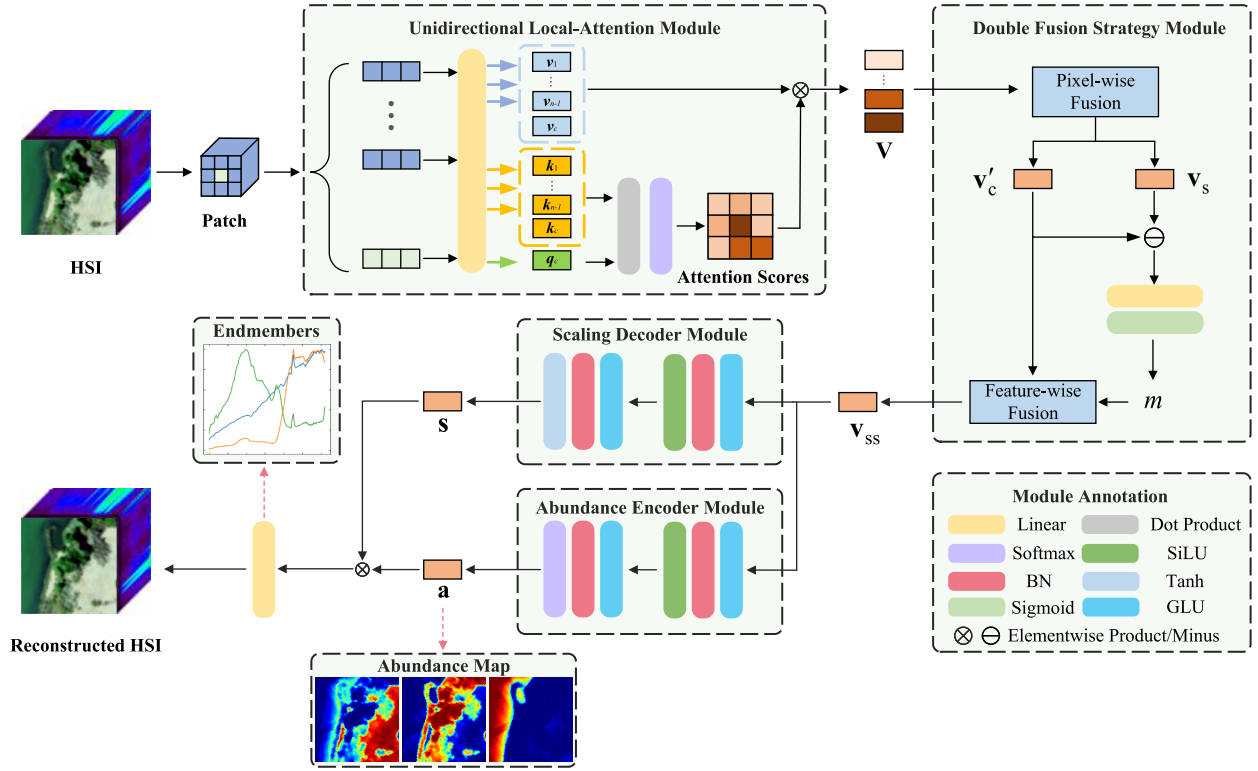


Fig. 1. Architecture of the proposed framework (ULA-Net) that consists of three cascaded modules. The first unidirectional local-attention (ULA) module aims to effectively explore local information, where the intensity of attention weights represents the significance of the corresponding data and V represents the weighted value vectors. The DFS is designed to realize spatial-spectral feature fusion and achieve precise control of the contribution of spatial features. Reconstructed HSI is obtained by a linear layer with a scaling decoder module.

The architecture of AEs can be summarized into two components: the encoder and the decoder. The input data is first encoded into a latent representation by the encoder, and then the decoder reconstructs the original input from this latent representation [42], [43], [44]. Assuming the AE is trained with batches of size N , this process can be formulated as

$$f_{\text{encode}}(\mathbf{Y}^T | \Theta_e) = \mathbf{A}^T \quad (3)$$

$$f_{\text{decode}}(\mathbf{A}^T | \Theta_d) = \hat{\mathbf{Y}}^T \quad (4)$$

where $f_{\text{encode}}(\cdot)$ and $f_{\text{decode}}(\cdot)$ denote the encoder network and decoder network, respectively. The abundance matrix $\mathbf{A}$ is the output of the encoder, while the endmember matrix $\mathbf{M}$ is not explicitly expressed in (4). The weights corresponding to the endmembers are generally contained in the parameters of the decoder network Θ_d (i.e., $\mathbf{M} \in \Theta_d$). Regarding the specific architectures of the encoder and decoder, various operators in neural networks can be utilized, such as fully connected layers, 1-D convolutions, 2-D convolutions, or 3-D convolutions, depending on the input data. In addition, the parameters of the AE Θ_e and Θ_d are learned by minimizing the loss function, which aims to make the input and reconstructed data as close as possible. To introduce nonlinearity, activation functions like Sigmoid, ReLU, and Tanh are commonly adopted in the AE.

C. Gated Linear Unit

The GLU [45] can be viewed as a variant of the linear layer by introducing a gating mechanism to control the flow of information. It consists of two parts: the first part applies

a linear transformation to the input data, and the second part applies a gating mechanism on the output of the linear transformation, controlling the proportion of outputs that can be passed to the next layer.

The gating mechanism in GLU is implemented using the Sigmoid activation function. The Sigmoid function maps the output of the linear transformation to a value between 0 and 1, which is then used to control the amount of information flowing through. Specifically, the output of GLU is given by

$$\text{GLU}(x) = (\mathbf{W}x + b) * \sigma(\mathbf{V}x + c) \quad (5)$$

where x denotes the input data, $\mathbf{W}$ and b are the learnable parameters of the linear transformation, $*$ represents element-wise multiplication, and σ represents the Sigmoid function.

III. PROPOSED METHOD

This section presents an introduction to our proposed network, which consists of a ULA module, a DFS module, and a scaling decoder. The architecture of our work is shown in Fig. 1. We introduce a ULA module to evaluate the significance of neighboring pixels, providing weighted output values. Following this, the DFS module performs pixel-wise and feature-wise fusion. The network then processes these fused features to generate scaling factors and abundance vectors. Finally, the reconstruction is accomplished through a single-layer linear operation.

A. ULA Mechanism

The multihead self-attention mechanism in transformer [29] provides an efficient attention mechanism that allows the model to jointly attend to different positions and representational subspaces of the input sequence. In addition, the multihead self-attention performs linear projections of the input into different spaces to produce multiple attention heads, each of which focuses on a different subspace. This enhances the model's capacity and generalization ability. Inspired by the multihead attention, we propose a ULA module. A schematic is shown in Fig. 1.

This module is a component of the encoder, tasked with extracting local homogeneity features from the input data. As shown in Fig. 1, the input is an image patch (a 3×3 patch size is used here as an example and can be extended in practice), where the green central pixel is the target pixel to be unmixed, and the blue surrounding pixels are the neighborhood pixels that provide spatial contextual information. For the central pixel, the ULA module first applies three linear transformations to obtain $\mathbf{q}_c$, $\mathbf{v}_c$ and $\mathbf{k}_c$, as follows:

$$\begin{aligned}\mathbf{q}_c &= \mathbf{x}_c \mathbf{W}^Q \\ \mathbf{k}_c &= \mathbf{x}_c \mathbf{W}^K \\ \mathbf{v}_c &= \mathbf{x}_c \mathbf{W}^V\end{aligned}\quad (6)$$

where the subscript c denotes the central pixel, $\mathbf{x}_c \in \mathbb{R}^{1 \times L}$ is the input central pixel, and $\mathbf{W}^Q$, $\mathbf{W}^K$, and $\mathbf{W}^V \in \mathbb{R}^{L \times d_{\text{model}}}$ are weight matrices of three fully connected layers, to project the input vectors to the model's hidden dimension d_{model} . The three linear transforms serve different purposes: $\mathbf{q}_c \in \mathbb{R}^{1 \times d_{\text{model}}}$ is the query vector, which serves as a reference to compare and assess the relevance of contextual information. $\mathbf{k}_c \in \mathbb{R}^{1 \times d_{\text{model}}}$ is the key vector for the central pixel itself. The query vector can match against its own key to increase the query weight and reduce the utilization of spatial context when neighbors provide less helpful information. $\mathbf{v}_c \in \mathbb{R}^{1 \times d_{\text{model}}}$ is the value vector of the central pixel, which can be seen as the projection or feature representation of the input in a subspace.

Similarly, each neighborhood pixel also undergoes two linear transformations to obtain $\mathbf{k}_l$ and $\mathbf{v}_l$, as follows:

$$\begin{aligned}\mathbf{k}_l &= \mathbf{x}_l \mathbf{W}^K \\ \mathbf{v}_l &= \mathbf{x}_l \mathbf{W}^V\end{aligned}\quad (7)$$

where $\mathbf{x}_l \in \mathbb{R}^{1 \times L}$ represents the l th neighborhood pixel (l is just an index in the set of neighboring pixels, not related to spatial order), $l = 1, 2, \dots, n-1$. $\mathbf{W}^K$ and $\mathbf{W}^V$ are shared with the central pixel. $\mathbf{k}_l \in \mathbb{R}^{1 \times d_{\text{model}}}$ and $\mathbf{v}_l \in \mathbb{R}^{1 \times d_{\text{model}}}$ are the key vector and value vector for the corresponding neighboring pixel, respectively. Note that, unlike the central pixel, the neighbors do not generate query vectors. This means that the neighboring pixels do not need to calculate their correlation with other pixels.

By computing the correlation between different neighboring pixels and the central pixel, distinct attention weights are assigned to them. This process can be seen as the central pixel searching for spectrally similar neighbors. Pixels with correlated spectral information will receive higher scores, allowing them to contribute more to the subsequent unmixing.

On the other hand, dissimilar pixels are expected to have query vector $\mathbf{q}_c$ and key vector $\mathbf{k}_l$ that are inverse to each other, i.e., their spectral angle is close to 180° . The attention scores are computed as follows:

$$\mathbf{t} = \text{Softmax}\left(\frac{\mathbf{q}_c \mathbf{K}^T}{\sqrt{d_{\text{model}}}}\right) \quad (8)$$

where $\mathbf{K} = [\mathbf{k}_c, \mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_{n-1}] \in \mathbb{R}^{n \times d_{\text{model}}}$ concatenates the key vectors of all pixels in the block, with n representing the number of pixels in a patch. The result of dot product $\mathbf{q}_c \mathbf{K}^T \in \mathbb{R}^{1 \times n}$ corresponds to the $n-1$ attention scores for the neighbors and one self-attention score for the central pixel. To normalize the unbounded attention scores, Softmax is applied to convert them into a probability distribution where each value lies in $[0, 1]$, representing the relative importance of each pixel. The denominator d_{model} in Softmax prevents numerical overflow/underflow and gradient vanishing issues during computation. Finally, the normalized attention scores form a vector $\mathbf{t} \in \mathbb{R}^{1 \times n}$.

After obtaining the attention scores, the value vectors are reweighted accordingly as follows:

$$\mathbf{v}'_l = t^{(l)} \cdot \mathbf{v}_l, \quad l = 1, 2, \dots, n-1 \quad (9)$$

$$\mathbf{v}'_c = t^{(c)} \cdot \mathbf{v}_c \quad (10)$$

where $\mathbf{v}' \in \mathbb{R}^{1 \times d_{\text{model}}}$ denotes the reweighted value vector and $t^{(l)}, t^{(c)} \in \mathbb{R}^1$ are scalar elements of the attention weight vector $\mathbf{t}$. In this process, important neighboring features are preserved due to their higher attention scores, while irrelevant spatial patterns are discarded as their weights approach zero. Consequently, we perform feature extraction on a pixel-by-pixel basis.

As a result, this module extracts valuable spatial information from neighboring pixels, which is then utilized in subsequent processing. Unlike the general self-attention mechanism in transformers, the proposed module computes correlations by projecting the target pixel to be unmixed as the query vector and assessing its correlation with other neighboring pixels. This approach avoids projecting neighboring pixels into query vectors and redundant computations of correlations between neighboring pixels. Information flow follows a unidirectional direction from the adjacent pixels toward the target pixel, so we call it a ULA mechanism.

B. Double Fusion Strategy

Based on the ULA module, we can obtain a weighted integration of the patch, where the neighboring pixels are assigned different weights according to their relevance to the central pixel. Thus, we propose the DFS module to achieve more flexible control of local homogeneous and unrelated information.

By aggregating the value vectors of all spatial pixels and the central pixel, we can perform pixel-wise information fusion and obtain the fused spatial feature, formulated as

$$\mathbf{v}_s = \mathbf{tV} = t^{(c)} \cdot \mathbf{v}_c + \sum_{l=1}^{n-1} t^{(l)} \cdot \mathbf{v}_l \quad (11)$$

where $\mathbf{v}_s$ denotes the aggregated spatial feature and $\mathbf{V} = [\mathbf{v}'_c, \mathbf{v}'_1, \mathbf{v}'_2, \dots, \mathbf{v}'_{n-1}] \in \mathbb{R}^{n \times d_{\text{model}}}$ is the matrix concatenating all value vectors. Notably, the spatial feature also includes the value representation of the central pixel. This accounts for cases when the center is an isolated point, e.g., surrounded by different materials, or the contextual information is irrelevant. The center can then increase its own attention weight to reduce the utilization of irrelevant spatial information adaptively.

Obtaining the fused spatial feature, we introduce the strategy for feature-wise fusion. For the spectral feature, we reuse the central value vector $\mathbf{v}'_c$ from the attention module to reduce model complexity. Inspired by [26], the fusion is performed by taking the extracted spatial and spectral features as references and letting the model learn the fusion weights. Differently, a subtraction between the two features is used here. The benefit of it is the ability to map the spatial and spectral representations into the same space. When they are close, the element-wise subtraction results in small values, providing the gating network with a salient input feature. Specifically, when the spatial context is similar to the spectrum, the attention weights and mask values will be small. With this characteristic of the mechanism, it is possible to control the contribution of spatial information. The fusion mechanism is formulated as

$$m = f_{\text{filter}}(\mathbf{v}_s - \mathbf{v}'_c | \Theta_g). \quad (12)$$

The filter represents a gating network, where Θ_g indicates the parameters of the network. The form of the network is a single fully connected layer, with the dimension changing from d_{model} to 1, and the activation function of the output is the Sigmoid function. Therefore, the gating factor $m \in (0, 1)$ is a scalar. According to (12), when spatial information is less informative and the output gating factor value is smaller, the distance between $\mathbf{v}_s$ and $\mathbf{v}'_c$ is more likely to be smaller. This represents that when extracting the spatial feature $\mathbf{v}_s$, the central pixel itself should receive the most significant attention weight. Specifically, in (11), $t^{(c)} \cdot \mathbf{v}_c$ should play a primary role, i.e., $t^{(c)}$ approaches 1, so that the generated $\mathbf{v}_s$ would be closer to $\mathbf{v}_c$.

By generating a spatial gating factor and performing multiplication with the spatial feature, followed by addition with the spectral feature, we obtain the feature-wise fused spatial and spectral features, as depicted below

$$\mathbf{v}_{ss} = \mathbf{v}'_c + m \cdot \mathbf{v}_s \quad (13)$$

$\mathbf{v}_{ss} \in \mathbb{R}^{1 \times d_{\text{model}}}$ represents the fused spatial-spectral feature. Based on pixel-wise and feature-wise spatial-spectral information fusion, the DFS module enables ULA-Net to achieve precise control over the weights assigned to local information, thereby regulating the contribution of each pixel. When local information is irrelevant to the central pixel, the ULA module assigns lower attention weights to surrounding pixels. DFS module accomplishes this by mapping the discrepancy between spatial and spectral features to small gating factors. Consequently, the weight of irrelevant information is further reduced, resulting in a sparse attention map.

C. Abundance Encoder

The goal of the ULA and DFS is to extract aggregated spatial and spectral features. After obtaining $\mathbf{v}_{ss}$, two feed-forward networks will produce an abundance vector and an endmember scaling vector separately. The feed-forward networks are structured with GLUs, with SiLU [46] as the activation function.

As for the abundance encoder, the input to the feed-forward networks is $\mathbf{v}_{ss} \in \mathbb{R}^{1 \times d_{\text{model}}}$, and then we use two layers of GLUs to obtain the abundance vector $\mathbf{a} \in \mathbb{R}^{1 \times P}$, where P is the number of endmembers. The process can be described as

$$\mathbf{a} = f_{\text{abundance}}(\mathbf{v}_{ss} | \Theta_a) \quad (14)$$

where $f_{\text{abundance}}(\cdot)$ is the abundance estimation network and Θ_a is the parameter of the network. The abundance estimation network uses the Softmax activation function for the output to obtain abundance values that sum to one and are nonnegative.

D. Spectral Scaling Decoder

Like the abundance encoder, the endmember scaling network uses two layers of GLUs to obtain the endmember scaling vector $\mathbf{s} \in \mathbb{R}^{1 \times P}$ from aggregated feature $\mathbf{v}_{ss}$. This process is described in the following equation:

$$\mathbf{s} = f_{\text{scale}}(\mathbf{v}_{ss} | \Theta_s) \quad (15)$$

where $f_{\text{scale}}(\cdot)$ is the endmember scaling estimation network with the parameter Θ_s . Different from the abundance encoder, the activation function used in the output layer of the endmember scaling network is the Tanh function. The scaling terms $\mathbf{s}$ are constrained to the range $(-1, 1)$, indicating the range of variation.

In real-world scenarios, the impact of spectral variability on imaging is limited. If the network generates scaling factors that are too large or too small, it can result in the loss of spectral features, rendering the outputs meaningless. Therefore, the magnitudes under meaningless cases can be limited by manually setting hyper-parameters, in the following manner:

$$\mathbf{S}_i = \text{Diag}(\mathbf{1}_P + \varepsilon \cdot \mathbf{s}_i) \quad (16)$$

$$\mathbf{M}'_i = \mathbf{M}\mathbf{S}_i \quad (17)$$

where $\cdot$ represents the scalar multiplication of a scalar and a vector. The unit vector $\mathbf{1}_P \in \mathbb{R}^{1 \times P}$ is a row vector, with each element being 1. $\mathbf{s}_i \in \mathbb{R}^{1 \times P}$ is the scaling vector generated by scaling decoder. Diag means transforming a row vector into the diagonal elements of a matrix. $\mathbf{S}_i \in \mathbb{R}^{P \times P}$ is a diagonal matrix with scaling factors for each endmember along the diagonal. And $\mathbf{M}'_i \in \mathbb{R}^{L \times P}$ represents the scaled endmembers in the i th pixel. The threshold ε is a manually set value that represents the maximum extent of the predefined endmember scaling, which takes values in the range of $(0, 1)$. In this way, the scaling range of the endmembers is limited to $(1 - \varepsilon, 1 + \varepsilon)$, $\mathbf{M}'$ is the endmembers after scaling, and $\mathbf{M}$ is the initial endmembers extracted by the preliminary endmember extraction method.

After scaling the endmembers, multiplying them by the abundance vector yields the reconstructed pixels, as shown

in (18). With this, the forward propagation process of the network is complete

$$\hat{\mathbf{x}} = \mathbf{M}'\mathbf{a}. \quad (18)$$

E. Loss Function

To overcome the inability of mean squared error (mse) to handle the spectral scaling effect and the parameter oscillation issue caused by the discontinuous gradient of spectral angle distance (SAD), we adopt the squared sine distance [47] as the reconstruction function

$$\mathcal{L}_{\text{re}} = \sin^2 \langle \hat{\mathbf{x}}, \mathbf{x} \rangle, \quad \text{s.t. } \langle \hat{\mathbf{x}}, \mathbf{x} \rangle < 90^\circ. \quad (19)$$

Since the abundance in AE networks is typically generated by the Softmax function, the resulting outputs tend to be dense due to the properties of the exponential function. To address the sparsity of land cover features in real-world scenarios, we employ the $L_{1/2}$ norm [48] to impose sparsity constraints

$$\mathcal{L}_{\text{sparsity}} = \alpha \|\mathbf{a}\|_{1/2} = \alpha \sum_{j=1}^P \sqrt{|a_j|}. \quad (20)$$

Due to the direct control of spatial information contribution and its impact on unmixing performance, it is necessary to add a constraint on the gating factor m to enhance its controllability and regulate its influence. Thus, we introduce a regularization constraint to prevent overfitting and enhance the model's robustness to noise

$$\mathcal{L}_{\text{gate}} = \beta * m. \quad (21)$$

The overall loss function of the proposed network is shown below

$$\mathcal{L}_{\text{all}} = \mathcal{L}_{\text{re}} + \mathcal{L}_{\text{sparsity}} + \mathcal{L}_{\text{gate}}. \quad (22)$$

IV. EXPERIMENTS

In this section, we present the experimental setup and results to evaluate the performance of our proposed method for effective unmixing. We conducted experiments on several public datasets and compared our method with state-of-the-art algorithms. The unmixing results are presented below.

A. Hyperspectral Dataset

1) *Synthetic Dataset*: The synthetic dataset was created by combining separate spectral and abundance data. The spectral data was sourced from the United States Geological Survey (USGS) mineral spectral library. The number of spectral bands was uniformly downsampled to 200. The abundance data was generated using a Gaussian field-based toolbox [49], satisfying nonnegativity and sum-to-one constraints. To consider spatial contextual correlations and model irregular lighting and terrain variations, the synthetic abundances were scaled and normalized using a 2-D Gaussian distribution with values ranging from 0.75 to 1.25. After synthesizing the HSIs, Gaussian-distributed random numbers were used to simulate noise. The format of the synthetic images was $70 \times 70 \times 200$, and a

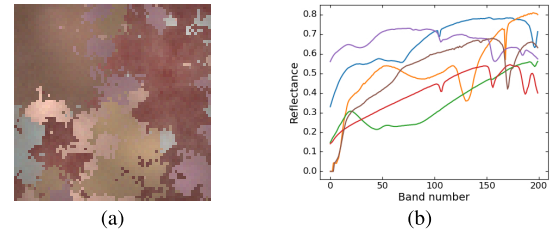


Fig. 2. Synthetic dataset (a) pseudo-color image and (b) endmembers.

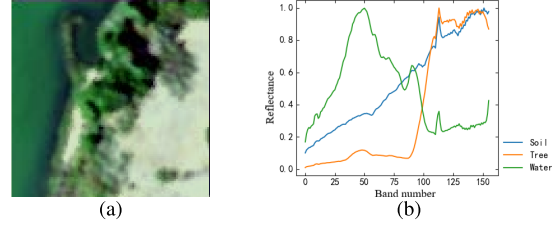


Fig. 3. Samson dataset (a) pseudo-color image and (b) endmembers.

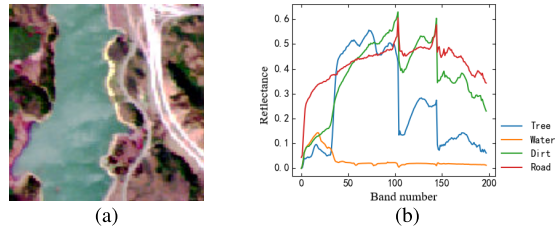


Fig. 4. Jasper Ridge dataset (a) pseudo-color image and (b) endmembers.

pseudocolor image, as shown in Fig. 2(a), was generated. The spectral curves used in the synthesis are depicted in Fig. 2(b)

$$\mathbf{Y} = \mathbf{M}\mathbf{A} + \mathbf{N}. \quad (23)$$

The mixing process is as described in (23), where $\mathbf{M}$ and $\mathbf{A}$ are the endmember matrix and the generated abundance matrix. $\mathbf{S}$ is a diagonal matrix with elements on the diagonal representing the scaling range for each pixel, and $\mathbf{N}$ represents simulated noise.

2) *Samson Dataset*: The pseudo-colored images of the Samson dataset [50] are shown in Fig. 3(a). The data is 95×95 pixels, with each pixel containing 156 channels. The wavelength range of the data is from 0.401 to 0.889 μm , with a spectral resolution of 3.13 nm. The image contains three different materials: soil, trees, and water. Their endmember spectra are shown in Fig. 3(b).

3) *Jasper Ridge Dataset*: The Jasper Ridge dataset [50] is shown in Fig. 4(a). The data consists of 100×100 pixels, with each pixel containing 224 bands. The wavelength range of the data is from 0.38 to 2.5 μm , with a spectral resolution of 9.46 nm. In addition, to mitigate the effects of water vapor absorption and atmospheric factors, bands 1–3, 108–112, 154–166, and 220–224 were removed, resulting in 198 remaining bands. The dataset contains four endmembers: trees, water, soil, and roads, as shown in Fig. 4(b).

B. Baselines and Metrics

1) *Baselines*: The six contrast algorithms used in this experiment are briefly introduced as follows:

- 1) *FCLS* [51]: The benchmarked algorithm for SU is used to establish a baseline for the unmixing results.
- 2) *MVCNMF* [52]: The nonnegative matrix factorization algorithm with the minimum simplex volume constraint is a linear unmixing algorithm based on the LMM.
- 3) *ALMM* [15]: Considered as one of the MLMs, this approach takes into account both spectral scaling factor and spectral perturbation, effectively addressing spectral variability.
- 4) *DCAE* [53]: The deep convolutional AE utilizes multiple 1-D convolutional layers to encode spectral information and reconstructs the data using a fully connected layer. However, this method does not take spatial information into account during unmixing.
- 5) *MACU* [54]: The non-LMM has been introduced into the AE, incorporating two branches in the decoder. These branches include a conventional linear mixing branch and a nonlinear branch. The final reconstruction is the sum of the outputs from both branches.
- 6) *GTCAN* [26]: The unmixing AE incorporates the gate mechanism and 3-D convolution to control the spatial information weight, achieved through a dual-branch network incorporating both spectral and spatial components.
- 7) *PGMSU* [38]: The probability generative model effectively addresses the issue of endmember variability by simultaneously generating endmembers and estimating abundance for each pixel. (To present the results of endmember estimation, we display the average curve of each endmember within every pixel in the following parts.)
- 8) *DeepTrans* [35]: By integrating the self-attention block within the latent space, the deep transformer unmixing model enhances its capability to leverage global information more effectively.

2) *Metrics*: When evaluating the performance of hyperspectral unmixing algorithms, multiple dimensions can be considered, including the model's accuracy of abundance estimation, and accuracy of endmember extraction. The effectiveness of abundance estimation is evaluated using the root mean square error of abundances (aRMSE)

$$\text{aRMSE} = \sqrt{\frac{1}{NP} \sum_{i=1}^N \sum_{j=1}^P (a_{ji} - \hat{a}_{ji})^2} \quad (24)$$

where a_{ji} denotes the true abundance of the j th endmember in the i th pixel, while $\hat{a}_{ji}$ represents the estimated abundance. N represents the number of pixels and P represents the number of endmembers. The effectiveness of endmember estimation is evaluated using the endmember SAD (eSAD)

$$\text{eSAD} = \frac{1}{P} \sum_{j=1}^P \arccos \left(\frac{\mathbf{m}_j^T \hat{\mathbf{m}}_j}{\|\mathbf{m}_j\|_2 \|\hat{\mathbf{m}}_j\|_2} \right) \quad (25)$$

where $\mathbf{m}_j$ represents the true spectrum of the j th endmember, while $\hat{\mathbf{m}}_j$ represents the estimated spectrum.

TABLE I

HYPERPARAMETER SETTINGS FOR SYNTHETIC AND REAL DATASETS

Parameter	α	β	ε	learning rate	epoch
Synthetic data	5e-3	1e-2	2e-1	6e-3	100
Real data	1e-1	1e-2	2e-1	6e-3	100

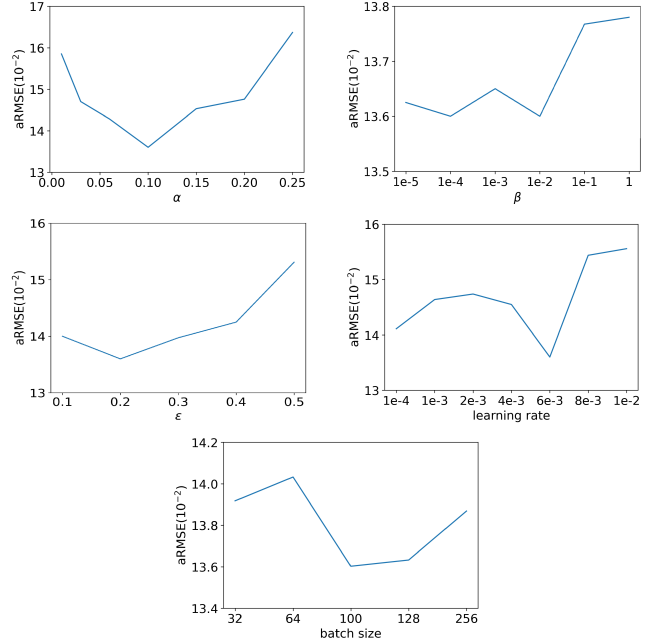


Fig. 5. RMSE as functions of the parameters α , β , ε , learning rate, and batch size.

C. Hyperparameters

The settings of hyperparameters have a significant impact on the experimental results. The following are the hyperparameter configurations used in this experiment.

The endmember preextraction method employed is VCA. To mitigate random noise, we conducted ten experiments and averaged the results. The hyperparameter settings are as follows: the gating openness penalty term β is set to 1e-2 and the spectral scaling factor ε is set to 2e-1. The optimizer used is AdamW [55], with an initial learning rate of 6e-3. The maximum number of iterations is set to 100, and the learning rate is updated by a factor of 0.99 every 20 iterations. The batch size is set to 100. In addition, the abundance sparsity constraints α vary across different datasets. We set α to 5e-3 and 1e-1 for synthetic and real datasets, respectively. To be more explicit, the hyperparameter settings are shown in Table I.

We conduct parameter analysis on the Jasper Ridge dataset. Fig. 5 illustrates the sensitivity of the proposed model to parameters such as α , β , m , learning rate, and batch size. Based on the results, the model performs well as hyperparameters vary within a reasonable range. Furthermore, it performs optimally at the parameter values selected for the experiments.

D. Experiment on Synthetic Dataset

1) *Quantitative Analysis*: The quantitative results on the synthetic dataset are shown in Table II. All AE-based

TABLE II
QUANTITATIVE RESULTS ON SYNTHETIC DATASETS (SNR = 30 dB). THE BEST PERFORMANCES ARE SHOWN IN BOLD

Methods	VCA-FCLS	MVCNMF	ALMM	DCAE	MACU	GTCAN	PGMSU	DeepTrans	Proposed
aRMSE	0.1046	0.1067	0.038	0.0427	0.1346	0.0348	0.1032	0.0449	0.0306
eSAD	0.0197	0.0222	0.0197	0.0159	0.0387	0.0158	0.0368	0.0168	0.0165

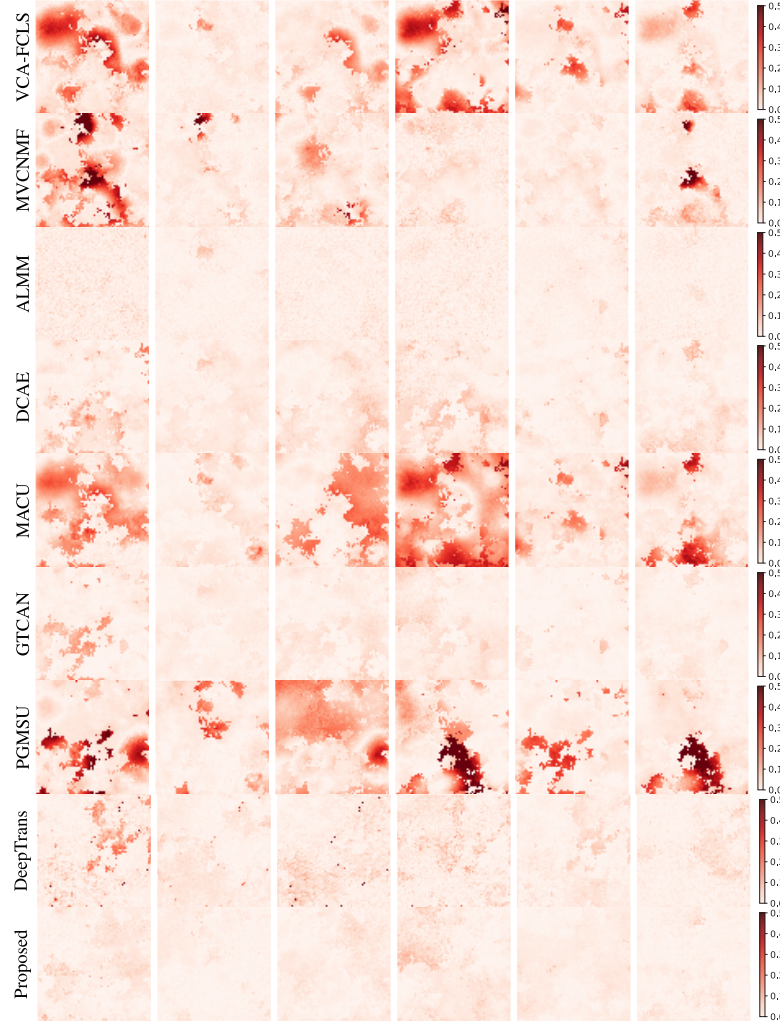


Fig. 6. Absolute abundance error of different algorithms on the simulated dataset.

algorithms utilize a single-layer decoder for endmember estimation, resulting in similar accuracy across these algorithms. Clearly, our work has achieved the best performance in abundance estimation. Experimental results demonstrate that by fully exploring spatial information and suppressing irrelevant information, the algorithm effectively mitigates the impact of noise, leading to higher accuracy in abundance estimation. Moreover, the network has also demonstrated good performance in endmember extraction.

2) *Visualization Analysis*: The abundance error distribution maps of different algorithms on the synthetic dataset are shown in Fig. 6. In the error distribution map, the intensity of color represents the magnitude of the error and lighter color indicates better performance. FCLS, MVC-NMF, DCAE, MACU, and PGMSU exhibit more noticeable abundance estimation errors. When observing the results of ALMM, particularly

in the first and fourth columns, there is slight noise present in the absolute abundance error map. These noise patterns are spatially discontinuous and uniformly distributed across the entire space. This is likely attributed to the presence of Gaussian noise added to the simulated data. ALMM, which does not incorporate spatial information, is unable to handle the noise effectively. However, GTCAN and DeepTrans do not encounter this issue as the introduction of spatial information results in correlated abundance estimates within neighborhoods, thus avoiding such errors. The main drawback of GTCAN and DeepTrans is its less accurate and flexible modeling of spectral variability, which fails to address the issue of inconsistent scaling among different endmembers within the same pixel. This leads to deviations in the abundance estimation of the first endmember. ULA demonstrates the best overall performance, consistent with the results presented in Table II.

TABLE III
ABUNDANCE RMSE RESULTS ON SAMSON DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

	VCA-FCLS	MVCNMF	ALMM	DCAE	MACU	GTCAN	PGMSU	DeepTrans	Proposed
Soil	0.2250	0.2247	0.1009	0.1883	0.3063	0.1109	0.2064	0.1090	0.0934
Tree	0.1813	0.1820	0.0548	0.0720	0.2971	0.0618	0.1825	0.0869	0.0537
Water	0.3388	0.3394	0.1064	0.1878	0.4300	0.1264	0.2838	0.2540	0.1039
Average	0.2484	0.2487	0.0874	0.1494	0.3445	0.0997	0.2242	0.0912	0.0837

TABLE IV
ENDMEMBER SAD RESULTS ON SAMSON DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

	VCA-FCLS	MVCNMF	ALMM	DCAE	MACU	GTCAN	PGMSU	DeepTrans	Proposed
Soil	0.0365	0.0239	0.0365	0.0288	0.0374	0.0140	0.0307	0.0196	0.0207
Tree	0.0309	0.0344	0.0309	0.0446	0.0305	0.0388	0.0560	0.0436	0.0289
Water	0.1245	0.1667	0.1245	0.1600	0.1321	0.1303	0.2096	0.1096	0.1206
Average	0.0640	0.0750	0.0640	0.0778	0.0667	0.0610	0.0987	0.0576	0.0567

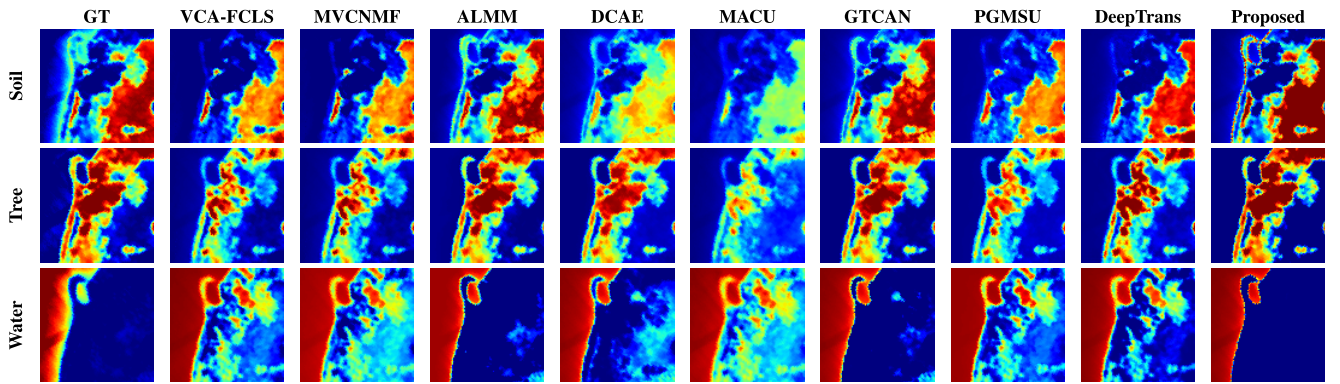


Fig. 7. Abundance map of Samson dataset.

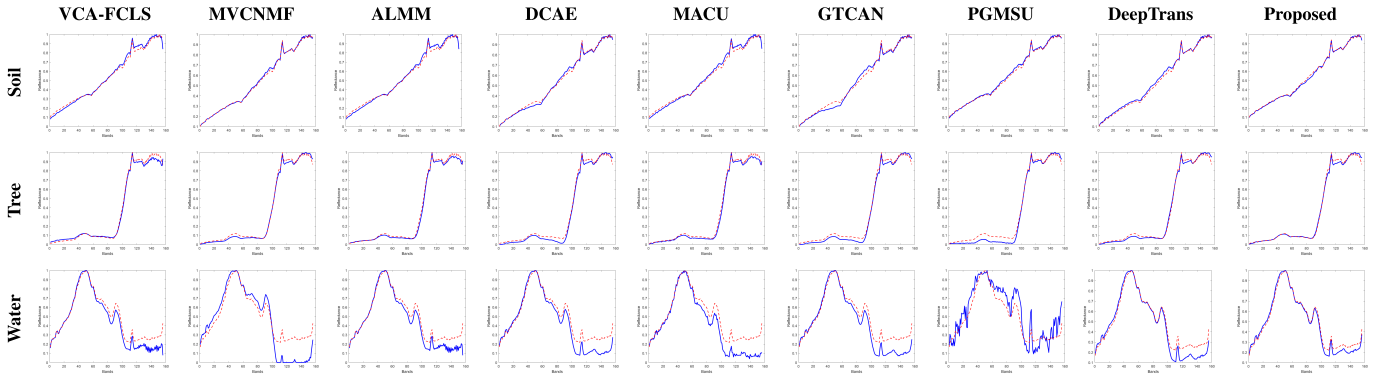


Fig. 8. Endmember curve of Samson dataset, where the red line is the true curve, and the blue line is the extraction results of different methods.

E. Experiment on Real Dataset

1) *Samson Dataset*: Our abundance and endmember estimation results for the Samson dataset are shown in Tables III and IV, respectively. The algorithms considering spectral variability, namely, ALMM and the proposed algorithm, achieved significantly better results. In addition, our approach fine-tuned the scaling term of ALMM, eliminated the perturbation term to reduce the algorithm's parameter count, and employed AEs for unmixing, yielding improved accuracy. By employing both spectral and spatial information and leveraging gating mechanisms to balance information, both our

approach and GTCAN achieved commendable performance. However, our proposed algorithm further improved the spectral information fusion strategy and considered spectral variability, leading to higher accuracy.

The abundance estimation maps and endmember curves for the Samson dataset are depicted in Figs. 7 and 8. The proposed algorithm achieves higher accuracy in abundance estimation compared to traditional linear unmixing algorithms. Compared to suboptimal algorithms, namely, GTCAN and DeepTrans, our approach utilizes spatial contextual information, resulting in enhanced purity and spatial consistency. Meanwhile,

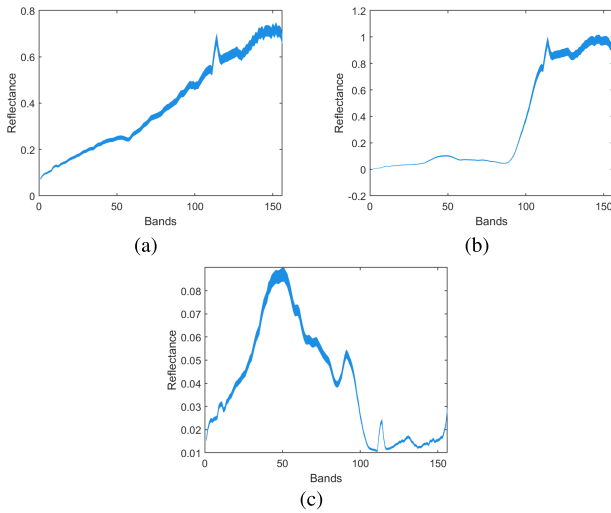


Fig. 9. Scaled endmembers of Samson dataset generated by the proposed model (a) scaled curve of endmember soil, (b) scaled curve of endmember tree, and (c) scaled curve of endmember water.

the ULA model leverages the contextual spatial data in a unidirectional-local-attention form, avoiding redundancy induced by global attention. For instance, in the abundance map of water, the land portion exhibits significantly better purity compared to other algorithms. Furthermore, our work demonstrates higher contrast in regions with complex land cover distributions. To better demonstrate our model's ability to model the spectral variability, the scaled endmember curves are shown in Fig. 9.

2) *Jasper Ridge Dataset*: The abundance maps and endmember curves for the Jasper Ridge Dataset are shown in Figs. 10 and 11. Similarly, our approach demonstrates high unmixing accuracy in regions with complex land cover distributions. Compared to other algorithms, the proposed method achieves superior purity, particularly evident in the abundance map of water. Moreover, by incorporating the ULA module, which offers spatial contextual information, as well as the DFS, our algorithm achieves enhanced accuracy in road recognition. Compared with the attention-based method DeepTrans, ULA also performs well. The global attention mechanism of DeepTrans successfully captures spatial information related to target pixels. However, the model faces challenges in discerning the influence of spectral variability in local regions, primarily attributed to some redundancy and misapplication of global information. In addition, other methods exhibit suboptimal performance due to the lack of mechanisms for information filtering or the suppression of irrelevant details. In regions with complex land cover distributions, we observe that the abundance map's boundaries are not sufficiently clear, leading to instances of misclassification. As a result, our abundance maps exhibit significant improvements compared to other algorithms. The quantitative results, as shown in Tables V and VI, also demonstrate the excellent performance of the proposed algorithm. Regarding endmember estimation, the model accurately identifies the characteristic peaks of endmembers, although its quantitative results slightly fall short of those achieved by DeepTrans. Also, the scaled endmember curves of the Samson dataset are shown in Fig. 12.

3) *Discussion*: To demonstrate the model's capability in handling complex conditions in real-world scenarios, we present scaled endmembers on real datasets, as illustrated in Figs. 9 and 12. When significant differences between the same materials arise due to variations in illumination, topography, and other factors, our proposed model can react to spectral variability differently for endmembers, generating distinct scaling factors and resulting in varied endmember curves in the figure. The scaling factors provide the model with parameters to characterize the spectral variability in the spectral mixing process, while also mitigating the impact on abundance estimation accuracy.

F. Sensitivity Analysis to SNR

The performance of various algorithms under different SNRs is illustrated in Fig. 13. The curve on the left represents the abundance estimation performance under different noise intensities. It can be observed that the proposed algorithm, along with DCAE and GTCAN, does not exhibit significant differences at low SNR. This may be attributed to the challenge of improving unmixing accuracy in the presence of high levels of noise. PGMSU demonstrates a higher unmixing accuracy, possibly attributed to its generation of endmembers for each pixel, which enables it to handle noise. As the SNR improves, the proposed algorithm demonstrates steady progress in accuracy, while DCAE and GTCAN show less pronounced improvements. This observation suggests that the improved spatial information utilization method possesses greater robustness, contributing to an enhanced upper limit of unmixing accuracy. Regarding the endmember results, all algorithms show similar performance without significant deviations. This is primarily because most of the decoders in the network consist of a single linear layer, and the weights of the decoder are initialized based on the vertex component analysis (VCA) algorithm. These weights are then iteratively updated under the guidance of the loss function. In the synthetic dataset, the spectral mixing patterns and noise effects on the dataset are simple. Therefore, the differences in performance regarding endmember extraction among various algorithms are insignificant.

G. Ablation Analysis

The proposed model consists of three crucial components: a ULA module, an endmember-wise scaling decoder, and a DFS module, which efficiently extract and fuse spatial and spectral features from hyperspectral data, effectively mitigating the interference caused by spectral variability and achieving higher accuracy in unmixing. To validate the effectiveness of the proposed modules, we conducted a series of ablation experiments to investigate the roles played by each module. Specifically, the ULA module serves as an encoder in the AE-based unmixing network, which is indispensable. Therefore, we utilize the structure of ResNet's BasicBlock and MaxPooling to substitute the ULA module, serving as the encoder. In addition, we examine the impact of the DFS and the scaling decoder on the unmixing performance. Experimental results on the Jasper Ridge dataset are presented in Table VII.

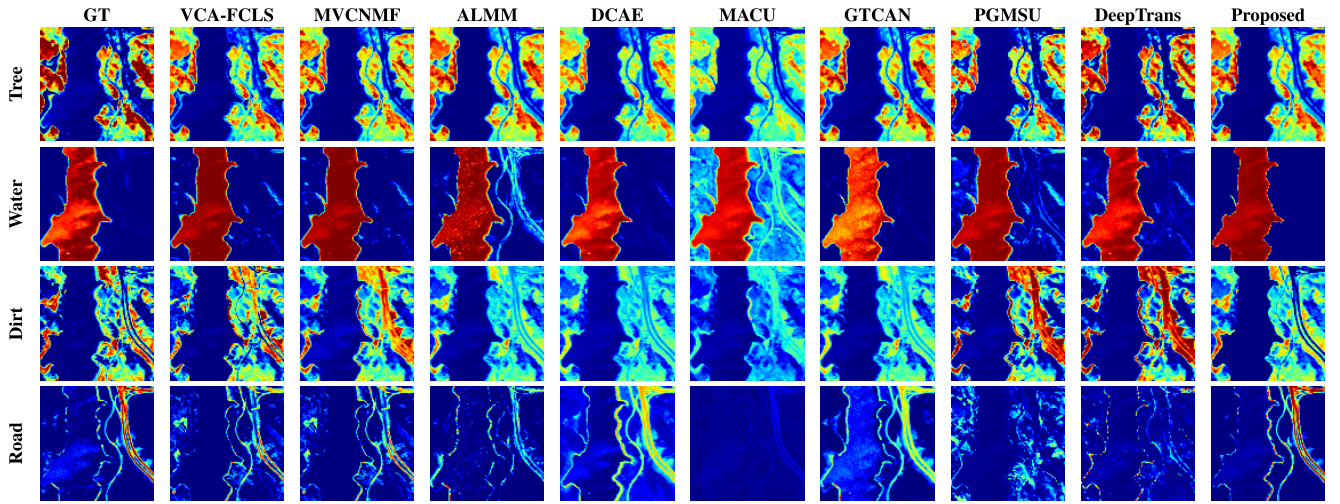


Fig. 10. Abundance map of Jasper Ridge dataset.

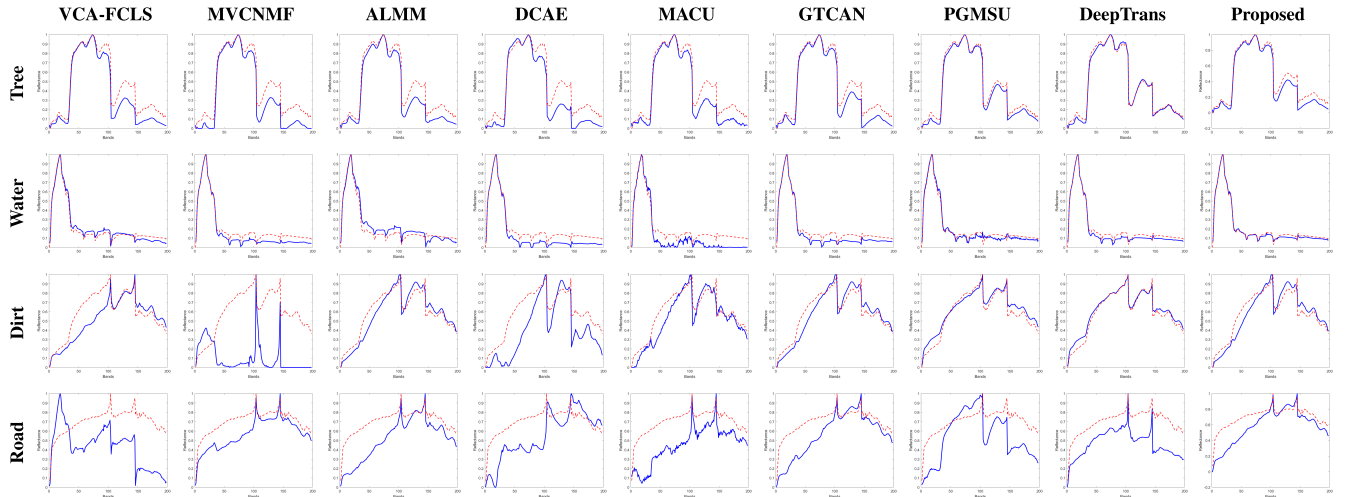


Fig. 11. Endmember curve of Jasper Ridge dataset, where the red line is the true curve, and the blue line is the extraction results of different methods.

TABLE V
ABUNDANCE RMSE RESULTS ON JASPER RIDGE DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

	VCA-FCLS	MVCNMF	ALMM	DCAE	MACU	GTCAN	PGMSU	DeepTrans	Proposed
Tree	0.1360	0.1357	0.1247	0.1383	0.2626	0.1193	0.1349	0.0889	0.1065
Water	0.0767	0.0779	0.1401	0.1723	0.2877	0.1498	0.1090	0.0911	0.1324
Dirt	0.2202	0.2188	0.1586	0.1796	0.2454	0.1446	0.2199	0.2843	0.1192
Road	0.2137	0.2102	0.1989	0.2152	0.2609	0.2051	0.2315	0.2937	0.1860
Average	0.1617	0.1607	0.1556	0.1764	0.2642	0.1547	0.1738	0.1895	0.1360

TABLE VI
ENDMEMBER SAD RESULTS ON JASPER RIDGE DATASETS. THE BEST PERFORMANCES ARE SHOWN IN BOLD

	VCA-FCLS	MVCNMF	ALMM	DCAE	MACU	GTCAN	PGMSU	DeepTrans	Proposed
Tree	0.2089	0.1898	0.2089	0.1731	0.2182	0.1573	0.1230	0.0453	0.1381
Water	0.1686	0.0975	0.1686	0.1698	0.2463	0.1555	0.1204	0.0582	0.0504
Dirt	0.2017	0.6004	0.2017	0.2271	0.2316	0.2004	0.1265	0.1318	0.1631
Road	0.4266	0.3014	0.4266	0.4792	0.4705	0.4393	0.4520	0.3825	0.3936
Overall	0.2515	0.2973	0.2515	0.2623	0.2917	0.2381	0.2055	0.1545	0.1863

By comparing the results of different experiments in Table VII, we can verify the effectiveness of the proposed modules. When performing unmixing w/o scaling

decoder (Case 1 compared to Case 3), the ULA module effectively extracts valuable spatial information and achieves higher precision in abundance estimation compared to the

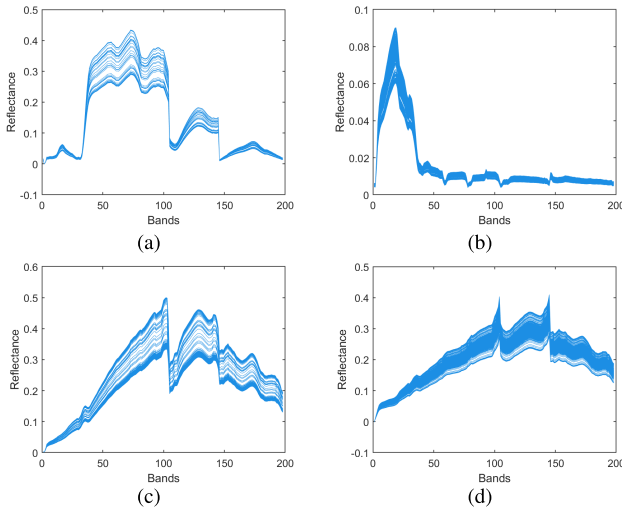


Fig. 12. Scaled endmembers of Jasper Ridge dataset generated by the proposed model (a) scaled curve of endmember tree, (b) scaled curve of endmember water, (c) scaled curve of endmember dirt, and (d) scaled curve of endmember road.

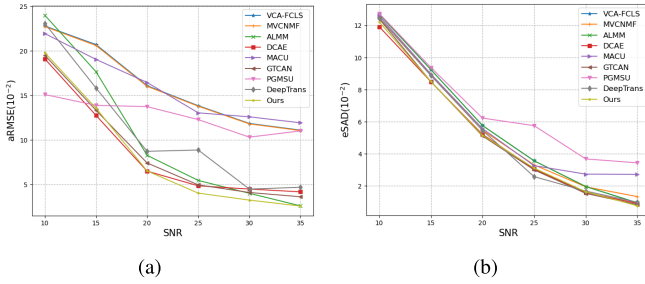


Fig. 13. Performance of algorithms under different SNR conditions (a) RMSE of abundance and (b) SAD of endmembers.

ResNet block. However, when considering spectral variability (Cases 2 and 5), the ResNet module fails to capture spectral variability information, resulting in a significant decrease in abundance estimation. In contrast, the ULA module incorporates efficient spatial extraction and dual-level spatial-spectral information fusion, enabling more flexible control over spatial information contributions. It provides valuable information for spectral variability unmixing models, leading to a significant improvement in abundance estimation compared to the w/o ULA cases, with the highest accuracy achieved.

Ablation studies on DFS demonstrate that by applying double spatial and spectral constraints on spatial information, the contribution of spatial information can be flexibly adjusted. Avoiding dense attention weights helps to weaken the impact of irrelevant spatial information on the accuracy of unmixing and achieves more accurate abundance estimation.

In addition, we conducted experiments to investigate the impact of ULA size on the unmixing results. The results in Table VIII indicate that our algorithm efficiently and accurately extracts and utilizes spatial information while mitigating the influence of redundant spatial information. Increasing the ULA size expands the receptive field, but due to the presence of local similarity, the added spatial information exhibits homogeneity. Thus, the network maintains a similar level of accuracy.

TABLE VII
QUANTITATIVE RESULTS ON ABLATION STUDY. THE BEST PERFORMANCES ARE SHOWN IN BOLD

Case	ULA	Scaling Decoder	DFS	aRMSE	eSAD
1	×	×	✓	0.1500	0.1873
2	×	✓	✓	0.1569	0.1861
3	✓	×	✓	0.1456	0.1897
4	✓	✓	×	0.1453	0.1866
5	✓	✓	✓	0.1360	0.1863

TABLE VIII
QUANTITATIVE RESULTS OF THE ABLATION STUDY ON THE SIZE OF ULA MODULE. THE BEST PERFORMANCES ARE SHOWN IN BOLD

ULA Size	aRMSE	eSAD	Time Cost(s)
3×3	13.60	18.63	75
5×5	13.56	19.08	77
7×7	13.71	18.88	80

In conclusion, the proposed model effectively suppresses the interference of spectral variability in unmixing, providing reliable information for abundance estimation and achieving higher unmixing precision.

V. CONCLUSION

In this article, we propose an SU network called ULA-Net that combines a spectral variability model with an AE. To extract helpful spatial and spectral information, we introduce the ULA module and the DFS module. ULA-Net achieves effective suppression of irrelevant spatial information through the DFS, enabling a better balance of weights between spectral and spatial information in the unmixing process. In addition, we introduce an endmember scaling module in the decoder, enhancing the model's flexibility by allowing different scaling factors for each endmember. In conclusion, our primary focus is on improving the stability and applicability of unmixing, and the advantages of the algorithm are demonstrated through rigorous experimentation.

REFERENCES

- [1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.
- [2] R. A. Borsoi et al., "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 223–270, Dec. 2021.
- [3] J. Chen, M. Zhao, X. Wang, C. Richard, and S. Rahardja, "Integration of physics-based and data-driven models for hyperspectral image unmixing: A summary of current methods," *IEEE Signal Process. Mag.*, vol. 40, no. 2, pp. 61–74, Mar. 2023.
- [4] Y. Shao and J. Lan, "A spectral unmixing method by maximum margin criterion and derivative weights to address spectral variability in hyperspectral imagery," *Remote Sens.*, vol. 11, no. 9, p. 1045, May 2019.
- [5] L. Drumetz, J. Chanussot, C. Jutten, W.-K. Ma, and A. Iwasaki, "Spectral variability aware blind hyperspectral image unmixing based on convex geometry," *IEEE Trans. Image Process.*, vol. 29, pp. 4568–4582, 2020.
- [6] A. Zare and K. C. Ho, "Endmember variability in hyperspectral analysis: Addressing spectral variability during spectral unmixing," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 95–104, Jan. 2014.

- [7] L. Ren, D. Hong, L. Gao, X. Sun, M. Huang, and J. Chanussot, "Orthogonal subspace unmixing to address spectral variability for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501713.
- [8] B. Somers, M. Zortea, A. Plaza, and G. P. Asner, "Automated extraction of image-based endmember bundles for improved spectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 396–408, Apr. 2012.
- [9] L. Drumetz, T. R. Meyer, J. Chanussot, A. L. Bertozzi, and C. Jutten, "Hyperspectral image unmixing with endmember bundles and group sparsity inducing mixed norms," *IEEE Trans. Image Process.*, vol. 28, no. 7, pp. 3435–3450, Jul. 2019.
- [10] T. Uezato, M. Fauvel, and N. Dobigeon, "Hyperspectral unmixing with spectral variability using adaptive bundles and double sparsity," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3980–3992, Jun. 2019.
- [11] D. Stein, "Application of the normal compositional model to the analysis of hyperspectral imagery," in *Proc. IEEE Workshop Adv. Techn. Anal. Remotely Sensed Data*, Greenbelt, MD, USA, Oct. 2003, pp. 44–51.
- [12] O. Eches, N. Dobigeon, C. Mailhes, and J.-Y. Tourneret, "Bayesian estimation of linear mixtures using the normal compositional model. Application to hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 19, no. 6, pp. 1403–1413, Jun. 2010.
- [13] X. Du, A. Zare, P. Gader, and D. Dranishnikov, "Spatial and spectral unmixing using the beta compositional model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 1994–2003, Jun. 2014.
- [14] P.-A. Thouvenin, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral unmixing with spectral variability using a perturbed linear mixing model," *IEEE Trans. Signal Process.*, vol. 64, no. 2, pp. 525–538, Feb. 2016.
- [15] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.
- [16] L. Drumetz, M.-A. Veganzones, S. Henrot, R. Phlypo, J. Chanussot, and C. Jutten, "Blind hyperspectral unmixing using an extended linear mixing model to address spectral variability," *IEEE Trans. Image Process.*, vol. 25, no. 8, pp. 3890–3905, Aug. 2016.
- [17] H. Liu, Y. Lu, Z. Wu, Q. Du, J. Chanussot, and Z. Wei, "Bayesian unmixing of hyperspectral image sequence with composite priors for abundance and endmember variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503515.
- [18] Y. Qu and H. Qi, "UDAS: An untied denoising autoencoder with sparsity for spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1698–1712, Mar. 2019.
- [19] R. A. Borsoi, T. Imbiriba, and J. C. M. Bermudez, "Deep generative end-member modeling: An application to unsupervised spectral unmixing," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 374–384, 2020.
- [20] Y. Su, J. Li, A. Plaza, A. Marinoni, P. Gamba, and S. Chakravorty, "DAEN: Deep autoencoder networks for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4309–4321, Jul. 2019.
- [21] G. A. Licciardi, X. Ceamanos, S. Douté, and J. Chanussot, "Unsupervised nonlinear spectral unmixing by means of NLPCA applied to hyperspectral imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Munich, Germany, Jul. 2012, pp. 1369–1372.
- [22] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503914.
- [23] B. Palsson, M. O. Ulfarsson, and J. R. Sveinsson, "Convolutional autoencoder for spectral-spatial hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 535–549, Jan. 2021.
- [24] D. Hong et al., "Endmember-guided unmixing network (EGU-Net): A general deep learning framework for self-supervised hyperspectral unmixing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6518–6531, Nov. 2022.
- [25] M. Zhao, S. Shi, J. Chen, and N. Dobigeon, "A 3-D-CNN framework for hyperspectral unmixing with spectral variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521914.
- [26] Z. Hua, X. Li, J. Jiang, and L. Zhao, "Gated autoencoder network for spectral-spatial hyperspectral unmixing," *Remote Sens.*, vol. 13, no. 16, p. 3147, Aug. 2021.
- [27] Z. Han, D. Hong, L. Gao, J. Yao, B. Zhang, and J. Chanussot, "Multimodal hyperspectral unmixing: Insights from attention networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524913.
- [28] L. Qi, X. Qin, F. Gao, J. Dong, and X. Gao, "SAWU-Net: Spatial attention weighted unmixing network for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, 2023, Art. no. 5505205.
- [29] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
- [30] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," Jun. 2021, *arXiv:2010.11929*.
- [31] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Montreal, QC, Canada, Oct. 2021, pp. 9992–10002.
- [32] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.
- [33] Y. Liu, X. Li, Z. Xu, and Z. Hua, "BSFormer: Transformer-based reconstruction network for hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, 2023, Art. no. 5507305.
- [34] J. Ding, X. Li, and L. Zhao, "CDFormer: A hyperspectral image change detection method based on transformer encoders," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6015405.
- [35] P. Ghosh, S. K. Roy, B. Koirala, B. Rasti, and P. Scheunders, "Hyperspectral unmixing using transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535116.
- [36] Y. Duan, X. Xu, T. Li, B. Pan, and Z. Shi, "UnDAT: Double-aware transformer for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522012.
- [37] Z. Yang, M. Xu, S. Liu, H. Sheng, and H. Zheng, "Spatial-spectral attention bilateral network for hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, 2023, Art. no. 5507505.
- [38] S. Shi, M. Zhao, L. Zhang, Y. Altmann, and J. Chen, "Probabilistic generative model for hyperspectral unmixing accounting for endmember variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516915.
- [39] M. Zhao, M. Wang, J. Chen, and S. Rahardja, "Hyperspectral unmixing for additive nonlinear models with a 3-D-CNN autoencoder network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509415.
- [40] L. Su, J. Liu, Y. Yuan, and Q. Chen, "A multi-attention autoencoder for hyperspectral unmixing based on the extended linear mixing model," *Remote Sens.*, vol. 15, no. 11, p. 2898, Jun. 2023.
- [41] X. Xu, X. Song, T. Li, Z. Shi, and B. Pan, "Deep autoencoder for hyperspectral unmixing via global-local smoothing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524216.
- [42] Z. Dou, K. Gao, X. Zhang, H. Wang, and J. Wang, "Blind hyperspectral unmixing using dual branch deep autoencoder with orthogonal sparse prior," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 2428–2432.
- [43] S. Shi, M. Zhao, L. Zhang, and J. Chen, "Variational autoencoders for hyperspectral unmixing with endmember variability," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2021, pp. 1875–1879.
- [44] Z. Zhao, D. Hu, H. Wang, and X. Yu, "Minimum distance constrained sparse autoencoder network for hyperspectral unmixing," *J. Appl. Remote Sens.*, vol. 14, no. 4, Oct. 2020, Art. no. 048501.
- [45] Y. N. Dauphin, A. Fan, M. Auli, and D. Grangier, "Language modeling with gated convolutional networks," in *Proc. 34th Int. Conf. Mach. Learn.*, Jul. 2017, pp. 933–941.
- [46] S. Elfving, E. Uchibe, and K. Doya, "Sigmoid-weighted linear units for neural network function approximation in reinforcement learning," *Neural Netw.*, vol. 107, pp. 3–11, Nov. 2018.
- [47] Z. Hua, X. Li, Y. Feng, and L. Zhao, "Dual branch autoencoder network for spectral-spatial hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 5507305.
- [48] Z. Xu, X. Chang, F. Xu, and H. Zhang, " $L_{1/2}$ regularization: A thresholding representation theory and a fast solver," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 23, no. 7, pp. 1013–1027, Jul. 2012.
- [49] S. Grupo de Inteligencia Computacional, Universidad del País Vasco/Euskal Herriko Unibertsitatea. *Hyperspectral Imagery Synthesis Toolbox*. Accessed: May 15, 2020. [Online]. Available: http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Imagery_Synthesis_tools_for_MATLAB
- [50] F. Zhu, Y. Wang, B. Fan, S. Xiang, G. Meng, and C. Pan, "Spectral unmixing via data-guided sparsity," *IEEE Trans. Image Process.*, vol. 23, no. 12, pp. 5412–5427, Dec. 2014.
- [51] D. C. Heinz, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.

- [52] L. Miao and H. Qi, "Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 3, pp. 765–777, Mar. 2007.
- [53] M. M. Elkholy, M. Mostafa, H. M. Ebied, and M. F. Tolba, "Hyperspectral unmixing using deep convolutional autoencoder," *Int. J. Remote Sens.*, vol. 41, no. 12, pp. 4799–4819, Jun. 2020, doi: [10.1080/01431161.2020.1724346](https://doi.org/10.1080/01431161.2020.1724346).
- [54] H. Li, R. A. Borsoi, T. Imbiriba, P. Closas, J. C. M. Bermudez, and D. Erdoğmuş, "Model-based deep autoencoder networks for nonlinear hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 5506105.
- [55] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," Jan. 2019, *arXiv:1711.05101*.



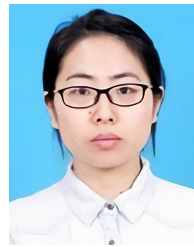
Shu Xiang received the B.S. degree from Zhejiang University, Hangzhou, China, in 2022, where he is currently pursuing the Ph.D. degree in control theory and control engineering.

His research interests include hyperspectral unmixing, image processing, and pattern recognition.



Jigang Ding received the B.S. degree from the School of Electrical Engineering, Beijing Jiaotong University, Beijing, China, in 2020. He is currently pursuing the Ph.D. degree in control theory and control engineering with Zhejiang University, Hangzhou, China.

His research interests include image processing and pattern recognition.



Shuhan Chen (Member, IEEE) received the B.S. degree from Ludong University, Yantai, China, in 2011, the M.S. degree from Liaoning Technical University, Huludao, China, in 2014, and the Ph.D. degree from Zhejiang University, Hangzhou, China, in 2020.

She did research as a Visiting Faculty Assistant at the Remote Sensing Signal and Image Processing Laboratory (RSSIPL), Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, from 2018 to 2020. She did a post-doctoral position at the Department of Control Science and Engineering, Zhejiang University, where she is currently an Assistant Professor at the Department of Control Science and Engineering. Her research interests include image registration, hyperspectral image processing, and pattern recognition.



Xiaorun Li received the B.S. degree from the National University of Defense Technology, Changsha, China, in 1992, and the M.S. and Ph.D. degrees from Zhejiang University, Hangzhou, China, in 1995 and 2008, respectively.

Since 1995, he has been with Zhejiang University, where he is currently a Professor with the College of Electrical Engineering. His research interests include hyperspectral image processing, signal and image processing, and pattern recognition.



Ziqiang Hua received the B.S. degree from Nanjing Normal University, Nanjing, China, in 2018, and the Ph.D. degree from Zhejiang University, Hangzhou, China, in 2023.

He is currently a Researcher at Zhejiang Laboratory, Hangzhou. His research interests include machine learning, hyperspectral image unmixing, and control theory.